

Subash Khanal, PhD

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🌐 Personal Website

🎓 Google Scholar

EDUCATION

Washington University in St. Louis

Doctor of Philosophy in Computer Science

St. Louis, MO, U.S.

2022 - 2025

University of Kentucky

Master of Science in Electrical Engineering (GPA: 4.0/4.0)

Lexington, KY, U.S.

2018 - 2020

Nitte Meenakhshi Institute of Technology

Bachelor in Electronics and Communication Engineering (GPA: 9.54/10 - Gold Medalist)

Bengaluru, India

2012 - 2016

WORKING EXPERIENCE

TerraByte AI, Inc.

Founding AI Scientist

August 2026 - present

- Contributing to the company's scientific proposals for various federal grants.
- Leading a project on FireTrack: Benchmarking Wildfire-State Tracking under Heterogeneous Earth Observation Coverage.
- Leading a project on multi-domain change detection of Earth's surface using multimodal Earth observation data.

Multimodal Vision Research Lab

Postdoctoral Research Associate

Washington University in St. Louis

August 2025 - August 2026

- Led development of a generative engine for hierarchical satellite image synthesis that completes full multi-resolution tile pyramids from sparse seed observations, while also achieving state-of-the-art performance on both super-resolution and outpainting subtasks. (**GENESIS**, SIGSPATIAL 2026)
- Co-led a systematic evaluation of geospatial foundation models for global field boundary delineation, and developed a U-Net-based approach with tailored losses and augmentations for improved accuracy and robustness across diverse conditions. (**PRUE**, CVPR 2026)
- Contributed to a deepfake detection framework that improves generalization to unseen generative models by enforcing real-image distribution alignment via latent blending, enabling robust, generator-agnostic detection. (**SimLBR**, CVPR 2026)
- Contributed to a probabilistic multimodal embedding model enabling any-to-any modality inference via masked reconstruction, supporting principled modality fusion and improved cross-modal retrieval and representation learning. (**ProM3E**, CVPR 2026)
- Mentored graduate students on multimodal learning research projects.

Digital Intelligence & Innovation Accelerator

Technical Lead

Washington University in St. Louis

February 2026 - August 2026

- Led development of an AI-driven disease trajectory model integrating genomic and longitudinal EHR data to identify synergistic comorbidity combinations and age-specific windows that accelerate Alzheimer's disease progression.
- Led development of a precision medicine dashboard visualizing cancer susceptibility and social determinants of health for the Coldwater Creek community in Missouri, enabling equitable access to environmental health data.
- Led development of an AI-guided computer vision system for objective, video-based facial rehabilitation assessment, transitioning recovery tracking from subjective observation to data-driven analysis.

Multimodal Vision Research Lab

Graduate Research Assistant

Washington University in St. Louis

August 2022 - August 2025

- Developed multimodal deep learning frameworks for geospatial understanding of global soundscapes.
 - **Sat2Sound** (EarthVision 2026): Developed a state-of-the-art soundscape mapping framework that leverages a Vision-Language Model (VLM) to enrich the semantic understanding of a location's soundscape, learns a shared codebook for fine-grained alignment, and enables retrieval-based, location-conditioned soundscape generation.
 - **PSM** (ACM Multimedia 2024): Extended GeoCLAP with a probabilistic, multi-scale, and metadata-aware embedding space for improved zero-shot soundscape mapping.
 - **GeoCLAP** (BMVC 2023): Proposed a tri-modal embedding space integrating satellite imagery, audio, and textual descriptions for zero-shot soundscape mapping at a global scale.
- Built large-scale ML-ready datasets:
 - **GeoSound**: A multi-resolution dataset combining over 300k paired samples of satellite imagery, geotagged sounds, and textual sound descriptions.

- **EarthCaps**: A global dataset comprising ~2.5 million satellite images and ~10 million auditory and visual captions, covering the entire Earth, generated using a state-of-the-art multimodal LLM.
- Contributed to **RANGE** (CVPR 2025), a novel retrieval-augmented method for multi-resolution geolocation embeddings.
- Contributed to additional multimodal embedding spaces:
 - **Sat2Cap** (EarthVision 2024) and **GeoBind** (IGARSS 2024) for query-driven mapping.
 - **TaxaBind** (WACV 2025) for ecological applications.
- Contributed to frameworks for ecological modeling and visual classification:
 - **LD-SDM** and **BirdSAT** (WACV 2024) for species distribution modeling and fine-grained classification.
- Contributed to diffusion model-based conditional generation methods:
 - **GeoSynth** (EarthVision 2024): A framework for controlled satellite image synthesis.
 - **MVPS** (TMLR 2025): A framework for novel panoramic view synthesis using satellite imagery.
- Collaborated on the IARPA-funded **SMART** project:
 - Developed large-scale self-supervised learning (SSL) methods, including Masked Autoencoders (MAE), for remote sensing applications such as semantic change detection.

Valuation and Market Dynamics

Applied Scientist - Intern

Zillow Group

May 2024 - August 2024

- Explored directions for improving Zillow's flagship product, Zestimate, by incorporating a diverse set of features.
- Developed a multimodal model for sales price estimation, learning from both structured tabular data and floorplan images.
- The proposed multimodal model achieved approximately 3% improvement across all metrics compared to the baseline.

Lin Brain Lab

Graduate Research Assistant

University of Kentucky

August 2020 - August 2022

- Provided applied Machine Learning (ML) and data science support to advance Alzheimer's disease (AD) research while working on different modalities such as medical imaging, electronic health records, and genomics.
- Designed CNN and Vision Transformers (ViT) based models trained on MRI/PET imagery for AD prediction.
- Focusing on interpretability, designed CAT-XPLAIN, an inherently interpretable ViT model that uses causality in predictions to identify the most important image patches.
- Built ML models trained on genetics, electronic health records, and imaging features for biomarker discovery and early prediction of Alzheimer's disease.

Speech and Signal Processing Lab

Graduate Research Assistant

University of Kentucky

August 2018 - August 2020

- Performed analysis of articulatory differences in the speech of native and non-native speakers of English.
- Built an Automatic Speech Recognition (ASR) based Mispronunciation Detection and Diagnosis (MDD) framework. ASR was trained using Recurrent Neural Networks (RNNs) on both articulatory and acoustic features.

Kantipur Engineering College

Lecturer

Lalitpur, Nepal

April 2017 - July 2018

- Taught courses: Microprocessor, Instrumentation
- Designed and conducted a lab on Digital Signal Processing (DSP), Microprocessor.

KEY SKILLS

- **Languages and Tools**: Agentic AI (experience), Python (fluent), PyTorch (fluent), Git (fluent), Docker (familiar), QGIS (familiar)
- **Research Interests and Relevant Courses**: Generative AI, Geospatial AI, Computer Vision, Multimodal Machine Learning, Deep Learning, Self-Supervised Learning, Data Science, Data Structures and Algorithms

AWARDS AND ACHIEVEMENTS

- COMPEX Scholarship offered by the Indian Embassy in Nepal for undergraduate study in India, 2012–2016.
- ECE Gold Medal, 2016 for graduating with Rank 1 in the department.

Preprints:

- Robinson Caleb, Muhawenayo Gedeon, **Khanal Subash**, Fang Zhanpei, Corley Isaac, Tàrano Ana M, Estes Lyndon, Marcus Jennifer, Jacobs Nathan, and Kerner Hannah, "The first global agricultural field boundary map at 10m resolution," arXiv:2605.11055, 2026.
- **Khanal Subash**, Brodie Benjamin, Xing Xin, Lin Ai-Ling, and Jacobs Nathan, "Causality for inherently explainable transformers: CAT-XPLAIN," **Spotlight Presentation** at XAI4CV: Explainable Artificial Intelligence for Computer Vision (CVPR Workshop), arXiv:2206.14841, 2022.

Accepted Papers:

- **Khanal Subash**, Cui Yangzhi, Cher Daniel, Xing Eric, Wei Brian, Sastry Srikumar, and Jacobs Nathan, "Genesis: A Generative Engine for Hierarchical Satellite Image Synthesis," ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL), 2026. **Application Track - Oral**.
- Sastry Srikumar, Cher Daniel, Wei Brian, Dhakal Aayush, **Khanal Subash**, Gupta Dev, and Jacobs Nathan, "TerraDiT: Point-Conditioned Diffusion Transformer for Satellite Image Synthesis," In: TerraBytes — Towards global datasets and models for Earth observation (European Conference on Computer Vision (ECCV) Workshop), 2026. **Best Paper Award**.

Published Papers:

- **Khanal Subash**, Sastry Srikumar, Dhakal Aayush, Ahmad Adeel, Stylianou Abby, and Jacobs Nathan, "Sat2Sound: A Unified Framework for Zero-Shot Soundscape Mapping," IEEE/ISPRS Workshop: Large Scale Computer Vision for Remote Sensing (EARTHVISION), 2026.
- Muhawenayo Gedeon*, Robinson Caleb*, **Khanal Subash***, Fang Zhanpei*, Corley Isaac, Wollam Alexander, Gao Tianyi, Strnad Leonard, Avery Ryan, Estes Lyndon, Tàrano Ana M., Jacobs Nathan, Kerner Hannah "PRUE: A Practical Recipe for Field Boundary Segmentation at Scale, " IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. ¹
- Dhakal Aayush, **Khanal Subash**, Sastry Srikumar, Arndt Jacob, Dias Philipe Ambrozio, Lunga Dalton, Jacobs Nathan, "SimLBR: Learning to Detect Fake Images by Learning to Detect Real Images, " IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- Sastry Srikumar, **Khanal Subash**, Dhakal Aayush, Lin Jiayu, Cher Dan, Jarosz Phoenix, Jacobs Nathan, "ProM3E: Probabilistic Masked MultiModal Embedding Model for Ecology, " IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- Ahmad Adeel, Sastry Srikumar, Dhakal Aayush, **Khanal Subash**, Levering Alex, Gilani Hammad, and Jacobs Nathan, "Canopy height mapping in the Western Himalayas, Pakistan: A deep learning approach using GEDI and Sentinel-2 fusion, " International Journal of Applied Earth Observation and Geoinformation, 2026.
- Sastry Srikumar, Dhakal Aayush, Xing Eric, **Khanal Subash**, and Jacobs Nathan, "Global and Local Entailment Learning for Natural World Imagery, " International Conference on Computer Vision (ICCV), 2025.
- Sastry Srikumar, Xin Xing, Dhakal Aayush, **Khanal Subash**, Ahmad Adeel, and Jacobs Nathan, "LD-SDM: Language-Driven Hierarchical Species Distribution Modeling," Computer Vision for Ecology (CV4E) Workshop, ICCV 2025. arXiv:2312.08334.
- Xiong Zhexiao, Xing Xin, Workman Scott, **Khanal Subash**, and Jacobs Nathan, "Mixed-View Panorama Synthesis using Geospatially Guided Diffusion, " Transactions on Machine Learning Research(TMLR), 2025.
- Dhakal Aayush, Sastry Srikumar, **Khanal Subash**, Ahmad Adeel, Xing Eric, and Jacobs Nathan, "RANGE: Retrieval Augmented Neural Fields for Multi-Resolution Geo-Embeddings, " IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
- Sastry Srikumar, **Khanal Subash**, Dhakal Aayush, Ahmad Adeel, and Jacobs Nathan, "TaxaBind: A Unified Embedding Space for Ecological Applications," IEEE Winter Conference on Applications of Computer Vision (WACV), 2025. **Oral Presentation**.
- **Khanal Subash**, Xing Eric, Sastry Srikumar, Dhakal Aayush, Xiong Zhexiao, Ahmad Adeel, and Jacobs Nathan, "PSM: Learning Probabilistic Embeddings for Multi-scale Zero-Shot Soundscape Mapping," ACM Multimedia, 2024, doi: 10.1145/3664647.3681620.
- Dhakal Aayush, Ahmad Adeel, **Khanal Subash**, Sastry Srikumar, Kerner Hannah, and Jacobs Nathan, "Sat2Cap: Mapping Fine-Grained Textual Descriptions from Satellite Images," IEEE/ISPRS Workshop: Large Scale Computer

¹* Equal Contribution

Vision for Remote Sensing (EARTHVISION), 2024. **Best Paper Award.**

- Sastry Srikumar, **Khanal Subash**, Dhakal Aayush, and Jacobs Nathan, "GeoSynth: Contextually-Aware High-Resolution Satellite Image Synthesis," IEEE/ISPRS Workshop: Large Scale Computer Vision for Remote Sensing (EARTHVISION), 2024.
- Dhakal Aayush, **Khanal Subash**, Sastry Srikumar, Ahmad Adeel, and Jacobs Nathan, "GeoBind: Binding text, image, and audio through satellite images," IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2024. **Oral Presentation.**
- Sastry Srikumar, **Khanal Subash**, Dhakal Aayush, Di Huang, and Jacobs Nathan, "BirdSAT: Cross-View Contrastive Masked Autoencoders for Bird Species Classification and Mapping," IEEE Winter Conference on Applications of Computer Vision (WACV), 2024.
- **Khanal Subash**, Sastry Srikumar, Dhakal Aayush, and Jacobs Nathan, "Learning Tri-modal Embeddings for Zero-Shot Soundscape Mapping," British Machine Vision Conference (BMVC), 2023.
- Xing Xin, Liang Gongbo, Zhang Yu, **Khanal Subash**, Lin Ai-Ling, and Jacobs Nathan, "Advit: Vision transformer on multi-modality pet images for alzheimer disease diagnosis," IEEE International Symposium on Biomedical Imaging (ISBI), 2022, doi: 10.1109/ISBI52829.2022.9761584.
- **Khanal Subash**, Chen Jin, Jacobs Nathan, and Lin Ai-Ling, "Alzheimer's Disease Classification Using Genetic Data," IEEE International Conference on Bioinformatics and Biomedicine (BIBM) Workshop, 2021, doi: 10.1109/BIBM52615.2021.9669730.
- Brodie Benjamin, **Khanal Subash**, Rafique Muhammad Usman, Greenwell Connor, and Jacobs Nathan, "Hierarchical Probabilistic Embeddings for Multi-View Image Classification," IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2021, doi: 10.1109/IGARSS47720.2021.9554405.
- **Khanal Subash**, Johnson Michael T., Soleymanpour Mohammad, and Bozorg Narjes, "Mispronunciation Detection and Diagnosis for Mandarin Accented English Speech," International Conference on Speech Technology and Human-Computer Dialogue (SpeD), 2021, doi: 10.1109/SpeD53181.2021.9587408.
- **Khanal Subash**, Johnson Michael T., and Bozorg Narjes. "Articulatory Comparison of L1 and L2 Speech for Mispronunciation Diagnosis," IEEE Spoken Language Technology Workshop (SLT), 2021, doi: 10.1109/SLT48900.2021.9383574.
- **Khanal Subash**, "Mispronunciation Detection and Diagnosis for Mandarin Accented English Speech," Theses and Dissertations–Electrical and Computer Engineering, 156, 2020, doi: 10.13023/etd.2020.340.